Viscous flow around a rotating cylindrical rod

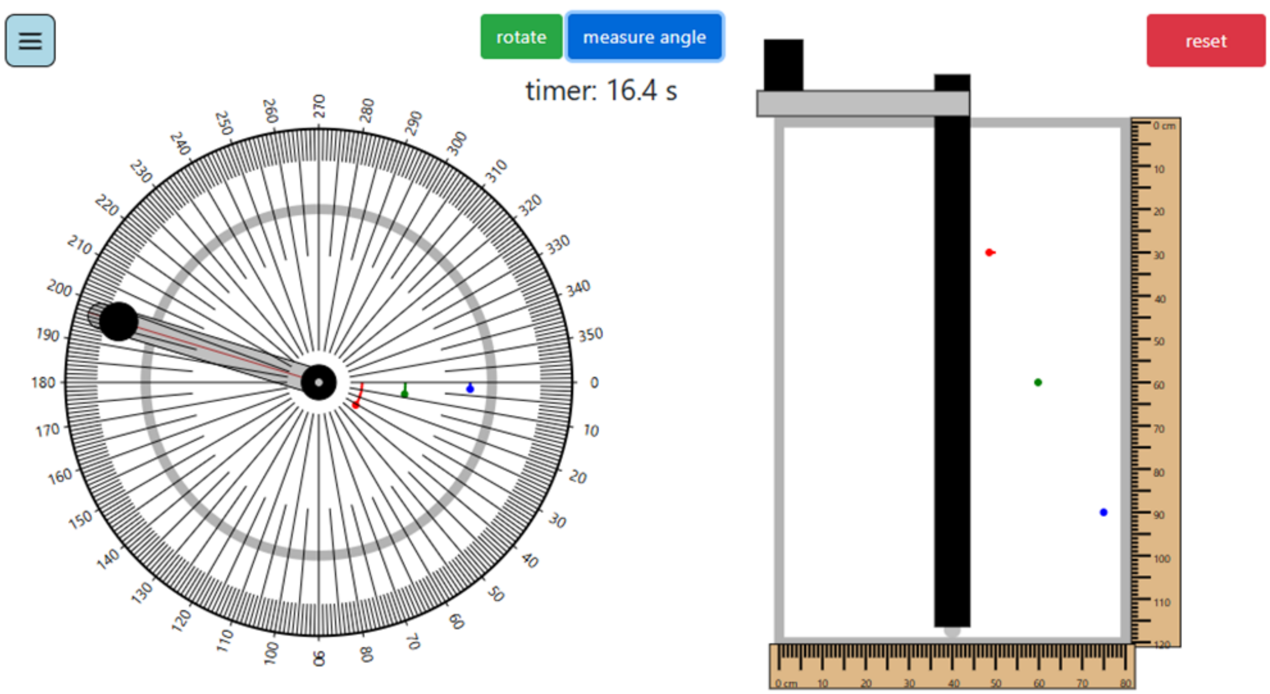
Name(s):____

The experiment consists of a cylindrical rod immersed in glycerol. The rod can rotate. Three dots represent three inks that stretch as the rod rotates. Their angular positions are tracked with time to measure the rotational velocity of each point. The goal of this experiment is to understand the velocity profile and compare it with prediction.

Student learning objectives

1. Measure velocity profile at low Reynolds number around a cylindrical rod.
2. Understand how velocity profile decays radially away from the rod.
3. Understand how velocity profile varies vertically for a long rod.
4. Derive the velocity profile for flow around a rotating rod.
5. Compare measured velocities with predictions.

Dimensions and details of the experiment



Click "rotate" to start the rod rotating clockwise and to also start the timer. Click "pause" to pause rotation and the timer. The "measure angle" button superimposes an angle scale on top of the cylinder. The rod can also be rotated manually by moving the handle on top of the cylinder on the left. The rod can be manually rotated either clockwise or counterclockwise.

Notation

v_θ = component of velocity in the θ direction

ω_{rod} = angular velocity of the rod

R_{rod} = the radius of the rod

r = the radial distance away from the center

$\Delta\theta_{point}$ = difference in angular position from the initial position for the points

$\Delta\theta_{rod}$ = difference in angular position from the initial position for the rod

Before starting the experiment.

1. If you rotate the rod clockwise, will the points move clockwise as well? Explain your reasoning.

2. If you rotate the rod by 360 degrees, how much will the red point rotate (less than, equal to, or greater than 360 degrees)? Explain your reasoning.

3. If you rotate the rod by 360 degrees, how much will the blue point rotate (less than, equal to, or greater than the red point)? Explain your reasoning.

4. Will gravity have any effect on the velocity profile? Explain your reasoning.

5. How many full rotations (i.e., in multiples of 360-degree rotations) of the rod will it take to make one full rotation of the blue point? Guess the answer and explain your reasoning.

6. If you rotate the rod three full rotations and rotate it in the opposite direction three full rotations such that it is back in its original location, what do you think will happen to the three points?

During the experiment

1. Measure the diameter of the cylindrical rod and the radial distance of the points away from the center of the rod. Move the ruler with the mouse and zoom with the scroll wheel to obtain an accurate measurement. Record the values in Table 1.

   Table 1 Radial distances

Measurement	Radial distance (cm)
Rod diameter (R_{rod})	
Red point (r_r)	
Green point (r_g)	
Blue point (r_b)	

2. Select “measure angle” to superimpose an angle scale. Click “rotate” to start the rod rotating slowly clockwise. The timer will start at the same time. Rotation will stop after four complete rotations (one full turn is 360 degrees).
3. Approximately every 15 seconds, click “pause” to pause the rotation and the timer. Record the time in Table 2. Hover your mouse over the angle scale (anywhere insider the circle) to display a line for the angle where the mouse pointer is. Use this to record angular positions of the rod and the three points in Table 2.
4. Repeat these measurements approximately every 15 seconds.
5. When the rod reaches 1440 degrees (four full rotations), rotation stops.

Table 2 Angular positions

	Angular position (degrees)			
Time (s)	rod	red	green	blue

6. Now rotate the cylinder slowly counterclockwise fully four times such that you are back in the original position. What do you observe about the position of the points compared to their original positions?

7. Plot angular positions versus time for the three points. On the same graph plot (rod angle)/4 so that all the curves fit on the same graph.

Analysis

1. Given that

$$v_\theta = \frac{\omega_{rod} R_{rod}^2}{r}$$

where v_θ is component of velocity in the θ direction
ω_{rod} is angular velocity of the rod
R_{rod} is the radius of the rod
r is the radial distance away from the center

suggest a plot to verify that your experiment aligns with this prediction. (*Hint:* $v_\theta = r\frac{d\,\theta}{dt}$). If you are unable to suggest a plot, consider plotting $\Delta\theta_{point} r_{point}^2 / R_{rod}$ versus $\Delta\theta_{rod}$, where $\Delta\theta_{point}$ and $\Delta\theta_{rod}$ represent the difference in angular positions from the initial position for the points and rod, respectively.

2. The density of glycerol is 1,260 kg/m^3 and its viscosity is 1,700 Pa·s. Define a Reynolds number for the experiment and calculate its value (*hint: calculate the angular velocity of the rod from the time it took to do one full turn*)

3. Why does gravity not impact the velocity profile? Is the pressure different at each point? If so, why does it not create a flow?

Here are two video links that demonstrate the viscous flow behavior:

https://www.youtube.com/watch?v=UpJ-kGII074

https://www.youtube.com/watch?v=j2_dJY_mIys